

Ethernet Standard and Specification

Task – 1 : Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.

Solution : RJ-45 Pin Layout and T568A / T568B

An **RJ-45 Ethernet connector** has **8 pins**. The two commonly used wiring standards are **T568A** and **T568B**.

RJ-45 Pin Layout

When viewing the connector with the **gold contacts facing you and the clip facing away/down**, the pins are numbered from **1 to 8** from left to right.

RJ-45 Connector Pin Layout and Wire Color Codes

RJ-45 Connector (Front View)



Pin 1 is on the left
(when the clip is facing down,
contacts facing you)

T568A Wiring Standard		
Pin	Color	Wire Illustration
1	White / Green	
2	Green	
3	White / Orange	
4	Blue	
5	White / Blue	
6	Orange	
7	White / Brown	
8	Brown	

T568B Wiring Standard		
Pin	Color	Wire Illustration
1	White / Orange	
2	Orange	
3	White / Green	
4	Blue	
5	White / Blue	
6	Green	
7	White / Brown	
8	Brown	

Task – 2 : Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Solution : **Ethernet Standards Comparison**

Ethernet Standard	Speed	Cable Type	Maximum Cable Length	Connector
10BASE-T	10 Mbps	Cat3 or better twisted-pair	100 m	RJ-45
100BASE-TX	100 Mbps	Cat5 or better twisted-pair	100 m	RJ-45
1000BASE-T	1 Gbps	Cat5e or better twisted-pair	100 m	RJ-45

Task – 3 : Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why. Hint : Think about how transmit and receive pairs are wired in each cable type.

Solution : Connecting Two Laptops Directly

If two laptops need to be connected **directly to each other without a switch or router**, traditionally a **crossover Ethernet cable** would be used.

In traditional Ethernet, the transmitting and receiving pairs of one device need to connect to the corresponding receiving and transmitting pairs of the other device.

➤ **Straight-Through vs Crossover**

Cable	Wiring	Traditional Use
Straight-through	Same standard at both ends, e.g. T568B → T568B	Laptop → Switch
Crossover	Different standards at each end, e.g. T568A → T568B	Laptop → Laptop

Important Modern Note

Most modern laptops and Ethernet devices support **Auto-MDI/MDI-X**, which automatically detects the transmit/receive configuration. Therefore, a **modern laptop-to-laptop connection may work with a normal straight-through cable as well.**

Task – 4 : List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

Solution : **Three Real-World Devices Using Ethernet**

1. Desktop Computer

A desktop computer can use Ethernet to connect to a **network switch or router**. A wired connection provides stable and reliable network communication.

Desktop PC ——— Ethernet Cable ——— Switch

2. Wi-Fi Router

A Wi-Fi router commonly has Ethernet ports that connect it to devices such as computers, switches, or an ISP's modem/ONT.

Internet/ONT ——— Router ——— Laptop
|
Ethernet

3. Network Printer

A network printer can use Ethernet to connect to the LAN. Multiple computers on the same network can then send print jobs to it.

Computer 1 —┐
Computer 2 —┤—— Switch —— Network Printer
Computer 3 —┐